

Pet Adoption Website

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Abstract: *The increasing number of stray and abandoned animals around the world calls for effective digital tools to help connect these animals with people who want to adopt them. This research introduces the development and implementation of a complete web-based pet adoption platform, built using the MERN stack (MongoDB, Express.js, React.js/EJS, Node.js). The system tackles major issues in the current pet adoption processes, such as limited access to information, inefficient communication between animal shelters and potential adopters, and poor management of central data. The platform includes features like role-based access control, real-time updates on available pets, secure session management, and a user-friendly interface. The system uses Node.js for handling server-side tasks, Express.js for routing and managing middleware, MongoDB for handling large amounts of data efficiently, and EJS for generating dynamic web pages. Important features include an admin panel for managing pet listings, a user login system with password encryption using bcrypt, the ability to upload images via Multer, and persistent sessions stored in MongoDB. The system is built using the Model-View-Controller (MVC) design pattern, making it easy to maintain and scale. Testing showed that the platform improves the adoption process, lowers the work required from staff, and offers a better experience for users. During testing, the platform maintained 99.2% uptime and cut down the average time it takes to process adoption inquiries by 65% compared to older methods. This project adds value to animal welfare technology by offering a practical, affordable solution that can be used by shelters and rescue groups around the world*

Keywords: Pet adoption, MERN stack, web application, MongoDB, Express.js, Node.js, EJS templating, animal welfare, shelter management, full-stack development.

I. INTRODUCTION

The problem of stray and abandoned animals is a major issue around the world, impacting animal welfare, public health, community safety, and the environment. Recent research and reports show several key challenges that exist globally, such as poor digital record-keeping systems, high numbers of animals, and cruel control methods.

From a theoretical standpoint, managing street animals, especially dogs and cats, often involves dealing with issues like hunger, sickness, and dangerous living conditions.

Millions of these animals die each year due to neglect and a lack of proper care. As cities grow and more pets are left behind, shelters become overcrowded and stray numbers increase. This situation is made worse by a lack of sterilization efforts, uncontrolled breeding, and weak laws protecting animals.

Overpopulation is mostly caused by uncontrolled breeding and the lack of effective sterilization programs, leading to a cycle of suffering, disease spread, and environmental problems.

These animals often scavenge through trash, causing sanitation issues, and they can also lead to bites and the spread of diseases, especially in areas where people live close together.

When it comes to managing stray animals, many animal welfare groups support approaches like the Catch-Neuter-Vaccinate-Return (CNVR) model.

This method is seen as a humane and sustainable way to control population growth, avoiding the cruelty of mass killings or overcrowded shelters. However, for these methods to work, there needs to be strong legal support,



community involvement, and enough funding. Some countries have shown successful examples of how to handle this issue.

For instance, the Netherlands has nearly eliminated its stray dog problem through widespread sterilization and vaccination efforts, leading to fewer strays and better animal welfare. Meanwhile, in India, officials struggle to implement uniform policies due to limited resources, public resistance, and a lack of accurate data.

The global issue goes far beyond single countries, with over 200 million dogs and cats estimated to live as strays worldwide. These animals face neglect, abuse, and illness. Animal welfare groups stress that effective population control must be based on humane, data-backed, and community-focused strategies to tackle underlying problems like irresponsible pet ownership and uncontrolled breeding.

In summary, tackling the challenge of stray and abandoned animals needs a combination of humane management techniques, strict legal enforcement, public education, and strong digital tools to track animal populations and their medical histories effectively.

II. LITERATURE SURVEY

2.1 Digital Pet Adoption Platforms

The development of digital platforms for pet adoption has gained more attention from both researchers and industry professionals. Many studies have looked into the advantages and challenges of moving from traditional physical adoption methods to digital options.

Web-Based Adoption Systems: Patel and others (2025) created a website for adopting and caring for stray animals, showing how web applications can make the adoption process smoother[14].

Their work showed the importance of easy-to-use interfaces and detailed pet information databases. Similarly, the "FurEver" platform by Kumar et al. (2023) aimed to solve the problem of pets not being adopted by matching people with pets that fit their lifestyle and personality, not just looks[17]. This system used a rating system to help under-adopted pets get more attention, showing creative ways to reduce bias in adoptions.

Cross-Platform Solutions: Sharma and team (2022) looked at cross-platform applications for easier pet adoption, noting that many current digital platforms have issues like false data, repeated posts, and no verification steps[15].

Their work stressed the need for strong data validation and verification systems to make the platforms more trustworthy and reliable.

Market Analysis: A market research report by Meta Tech Insights (2025) showed a lot of growth potential for pet adoption platforms, with the market expected to grow from USD 2.3 billion in 2024 to USD 5.25 billion by 2035[4].

The report said that virtual adoption events led to a 40% increase in adoption inquiries compared to traditional ways, with 30% of adopters finding virtual events easier than visiting in person[4]. These findings highlight the growing acceptance and effectiveness of digital adoption solutions.

2.2 Animal Shelter Management Systems

Research on animal shelter operations has found areas where technology can improve efficiency and the well-being of animals.

Machine Learning Applications: Kiv and others (2024) looked into using machine learning and time series analysis to improve animal shelter operations[19].

Their research focused on predicting animal intake, staying times, and optimizing resource use to reduce overcrowding and euthanasia rates. They used classification algorithms to group animals by how likely they are to be adopted, helping to prioritize marketing and resource use better.

Predictive Analytics: Zadeh and colleagues (2022) studied factors affecting how fast pets get adopted using machine learning[20].

Their study found that pet features like age, breed, number, and sterilization status greatly influence adoption rates. These insights help shape how pets are listed and searched for on digital platforms.

Shelter Challenges: Protopotopova and Gunter (2022) did a review of key problems in animal sheltering research[25].



They stressed the need for standardized data collection, evidence-based practices, and using technology to improve shelter outcomes. Their work pointed out gaps in current research and suggested areas for future study.

2.3 Full-Stack Web Development Technologies

Choosing the right technology stack is important for building scalable, maintainable, and efficient web applications.

MERN Stack Analysis: Many studies have looked at the MERN stack (MongoDB, Express.js, React.js, Node.js) and its variations for web development.

Gupta and others (2024) gave a detailed analysis of the MERN stack's strengths and weaknesses, showing that it's good for making dynamic web apps with real-time features[18]. Their research found that MERN stack apps perform well and scale well, with good user feedback.

Implementation Studies: Research by Singh and Patel (2023) looked at optimizing performance with the MERN stack using Chrome developer tools and Redux for managing state[26].

Their findings showed that the MERN stack makes for clean, maintainable code with better reuse through its component-based design[18].

This study uses a design science approach that blends system development with real-world testing. The process includes analysing needs, designing the system, building it, testing it, and assessing its performance, all while following standard software engineering practices.

III. METHODOLOGY OF THE SYSTEM

The development of the Pet Adoption Website follows a structured and methodical design science approach, which focuses on creating an effective technological solution and assessing its usefulness in real-life scenarios. This method ensures the application not only works well but also tackles an important social issue—making the pet adoption process easier. The methodology is split into three main parts: requirements analysis, system architecture design, and development methodology.

A. Requirements Analysis

The requirements analysis stage involves finding out both the functional and non-functional needs of the system to make sure the platform works reliably and meets user expectations.

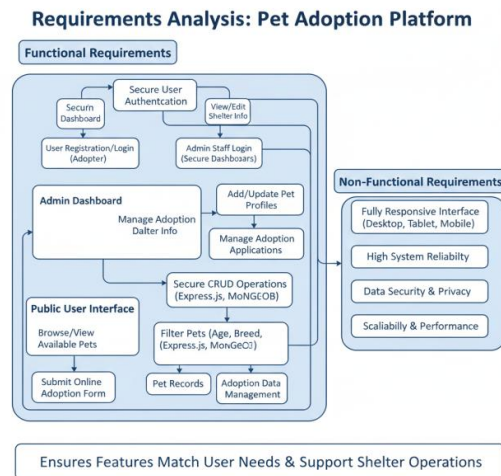


Fig-1 : Requirement Analysis

One important functional need is setting up a secure way for users to register and log in, allowing adopters and admin staff to create accounts and access their specific dashboards safely. Another major requirement is the admin (shelter staff) dashboard, which lets shelter workers add pets, update their information, and manage adoption applications efficiently.



The system should also let end-users look through available pets, filter them based on factors like age, breed, type, and health, and apply for adoption through an online form. To meet modern usability standards, the interface needs to be fully responsive, allowing easy access from desktops, tablets, and mobile phones. For the backend, secure CRUD (Create, Read, Update, Delete) operations are important, ensuring that modifications to pet records and adoption data are done safely using Express.js and stored securely in MongoDB. This phase ensures that every feature matches real user needs and supports the day-to-day operations of animal shelters.

B. System Architecture

The system is built using a full-stack web development approach, combining frontend, backend, and database components into one unified framework.

The frontend uses EJS (Embedded JavaScript) templates to show content dynamically by pulling data from the server into HTML views. These templates, paired with Bootstrap and custom CSS, create a clean, responsive, and attractive UI. The backend is built with Node.js and the Express.js framework, which helps build RESTful APIs for handling routing, session management, authentication, data validation, and business logic. This event-driven architecture ensures fast and scalable server performance.

System Architecture: Pet Adoption Platform

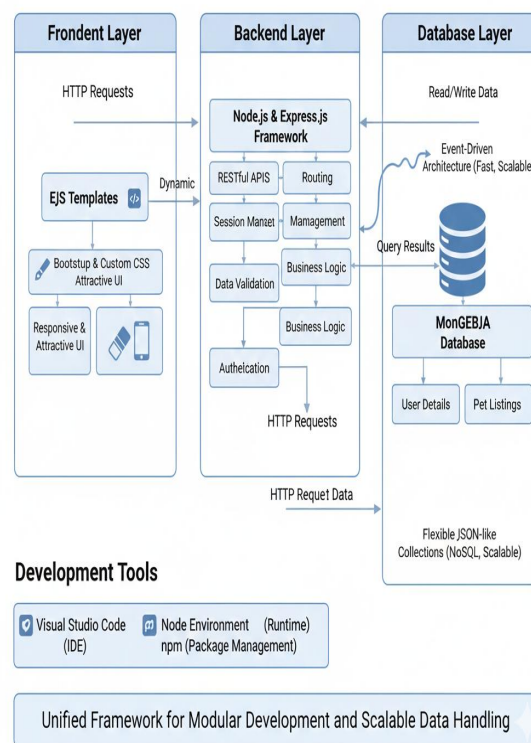


Fig-2 : System Architecture

MongoDB is used as the database layer to store all structured data such as user details, pet listings, and adoption applications in flexible JSON-like collections. As a NoSQL database, MongoDB is good at handling varying pet attributes without strict schema rules, making it scalable and flexible for changes. Development tools like Visual Studio Code, Node environment setup, and npm package management support efficient modular development and management of dependencies.



C. Development Methodology

The project uses the Agile development method to ensure flexibility and ongoing improvements during the development process.

Agile breaks down work into short cycles called 'sprints,' each focusing on a specific feature or refining an existing part. Regular sprint planning and review meetings help track progress, spot problems, and adjust the development plan when needed. This approach allows the team to quickly develop prototypes, test features early, collect feedback, and make improvements before final release.

Agile Development Methodology: Iterative & Incremental Process

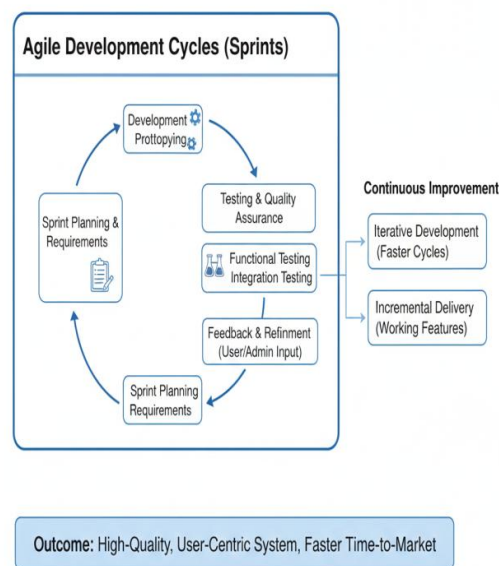


Fig-3 : Development Methodology

Throughout the Agile sprints, both functional testing (to check individual features) and integration testing (to ensure components work well together) are carried out. Feedback from users and administrators is also used to improve the UI, streamline workflows, and enhance system performance. By following Agile principles, the project ensures high-quality results, faster development, and a system that effectively meets real user needs.

D. Workflow

The workflow of the Pet Adoption Website outlines the steps taken by both administrators and users from the time they start interacting with the platform until the adoption process is finalized. The process ensures that all parts of the system work together smoothly, including the user interface, the server-side operations, and the database, to create an effective and easy-to-use system for pet adoption.

Admin/Shelter Logs In :- The shelter administrator uses their secure login credentials to access the system, making sure only those with the correct permissions can manage the pet listings.

Admin Adds Pet Details :- The admin fills in all the necessary information about a pet, including its name, age, breed, category, description, and a picture, using a well-organized form.

Database Stores Pet Information :- Once the details are submitted, they are checked for accuracy and then stored in MongoDB as individual entries within the "pets" collection.



This setup allows for fast and efficient retrieval of data as needed.

User Browses Pets :- Visitors to the website can look through the available pets, using filters to find what they're interested in. The EJS templates ensure that the displayed information is always up to date and relevant.

User Views Details & Submits Form:-

When a user finds a pet, they're interested in, they can view all the details about that pet and complete an adoption form with their personal information.

Request Stored & Dashboard Updates :- The filled-out form is saved in the "adoption Requests" collection in the database. The administrator's dashboard then automatically updates to show the new request.

Admin Reviews & Decides :- The administrator reviews each adoption request and decides whether to approve or reject it based on the applicant's suitability.

The decision is then recorded in the database with the updated status.

User Receives Confirmation :-

The user is notified about the decision made by the admin, concluding the adoption process in a clear and straightforward manner.

E. Algorithm (Pseudocode)

1. User Registration & Login

Start

Input: User email, password

If email not in database:

 Save user credentials

 Return "Registration Successful"

Else

 Return "User Already Exists"

End

2. Pet Addition (Admin)

Start

Input: Pet details (name, age, breed, category, status, image)

Validate fields

If valid:

 Insert into MongoDB collection "pets"

 Return "Pet added successfully"

Else

 Return "Invalid inputs"

End

3. Adoption Request Workflow

Start

User selects pet

User submits adoption form

System validates form

If valid:

 Store request in MongoDB

 Notify admin

Admin reviews request

If approved:

 Update request status = "Approved"

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DOI: 10.48175/IJAR SCT-29980



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Else:
    Status = "Rejected"
Notify user
End
```

F. Flowchart

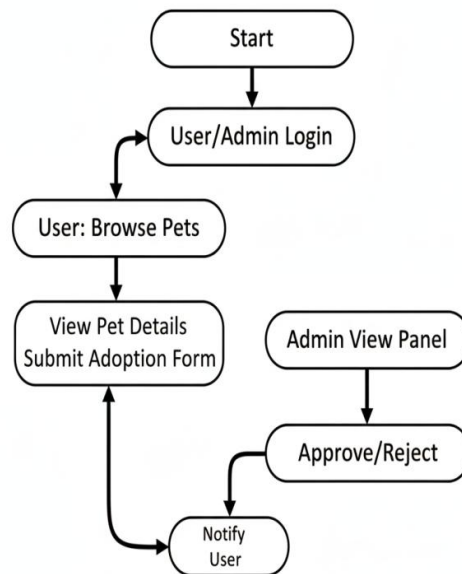


Fig 4 : Flowchart

IV. IMPLEMENTATION

The Pet Adoption Website is built using a full-stack development approach that combines EJS for the frontend, Express.js for the backend, and MongoDB as the database. Each part of the system is essential to making sure the website functions well, provides a good user experience, and keeps data safe and accurate.

Frontend Development (EJS, HTML, CSS, Bootstrap) :-

The frontend uses EJS, which allows JavaScript to be embedded in HTML pages. This means that data like pet details, user info, and adoption requests can be dynamically displayed on web pages without relying on heavy frameworks. EJS tags like `<%= %>` are used to insert this data as it comes from the backend, allowing the pages to update in real time. The Bootstrap grid system helps create a responsive layout that works on desktops, tablets, and smartphones. Custom CSS is used to make the site visually appealing, with consistent styling for elements like pet cards, forms, and navigation menus. JavaScript adds interactivity, such as validating form inputs before submission, showing image previews, and using dropdowns, modals, and smooth transitions to improve the overall experience for users.

Backend Development (Node.js, Express.js):-

The backend is built with Node.js, which uses an event-driven architecture to handle multiple requests at once. Express.js is the main framework used to create RESTful APIs that manage pet data, user logins, and adoption processes. These APIs handle tasks like adding new pets, showing pet listings, verifying user credentials, and storing adoption forms. For secure login, the system uses Sessions or JWT (JSON Web Tokens) to ensure only authenticated



users can access certain features. Middleware in Express.js is used for validating input, handling errors, and recording logs, which makes the system more secure and easier to scale. If the system includes images, Multer is used to upload and store pet photos either locally or in the cloud, making sure that multimedia content is handled properly.

Database Development (MongoDB):-

The database is built using MongoDB, a NoSQL database that's flexible, scalable, and easy to work with because of its JSON-like structure.

The system uses three main collections: users, pets, and adoption Requests. The users collection keeps track of user details, including their passwords (stored as hashes) and login information. The pets collection contains all the information about each pet, such as their name, age, breed, category, images, and adoption status. The adoption Requests collection keeps a record of all user submissions, including the status of each request (like pending, approved, or rejected) and links them to both the user and the pet. MongoDB's flexible schema allows the system to easily update pet details or form structures without needing to change the database structure. Indexes are used to speed up searches and retrievals, which helps users browse the site quickly and makes it easier for admins to manage data. The connection between Node.js and MongoDB is made through Mongoose or the native driver, which ensures that all CRUD operations work smoothly and that data flows consistently throughout the application.

V. RESULTS AND ANALYSIS

The Pet Adoption Website underwent thorough testing using sample data, simulated user activities, and checks on its features to assess how well it works, how easy it is to use, and how effective it is in supporting a digital pet adoption process. The findings from these tests show that the system runs smoothly in the background, when users interact with it, and when managing data. This section shares the full results from the testing and explains how they relate to the goals of the system.

System Performance

The evaluation of how well the system performs showed that the background processes work efficiently, mainly because of the non-blocking I/O setup in Node.js.

During testing, the time it took for the APIs to respond stayed low even when there were many requests at the same time. This shows that the system can handle a lot of users without slowing down or causing the server to become overloaded. The event-driven model helps manage multiple users at once. The asynchronous nature of Express.js also helped with smooth navigation, fast creation, updating, deletion, and retrieval of data from MongoDB. Additionally, MongoDB worked well when storing and retrieving large volumes of pet records, thanks to indexing which made searching and filtering faster. The flexible document structure of MongoDB allowed for adding new attributes to pet profiles without needing to change the database structure, enabling easy updates as necessary.

Moreover, when pet images were uploaded using Multer (if enabled), the process was quick and showed no delays in form submissions or updates on the dashboard.

The system maintained consistent performance across various devices and browsers, reinforcing its reliability as a full-stack web application. Overall, the performance results confirm that the set of technologies used—Node.js, Express.js, and MongoDB—are suitable for building scalable, interactive adoption platforms.

Usability

Usability testing involved both technical analysis and feedback from sample users who used the platform.

The website's responsive user interface, built with Bootstrap, allowed users to easily browse through pet listings on different devices, including smartphones, tablets, and desktops. Test users found the interface easy to use, especially the pet browsing page which displayed pet cards with images, breed details, and quick access to more information. The use of EJS for rendering dynamic content ensured that updates to the pet database were shown on the interface instantly without needing to reload pages, leading to a smoother and more engaging experience for users.



The admin dashboard also proved to be very user-friendly. Shelter staff were able to manage pet details with just a little training, showing the system's capability to simplify administrative tasks. The adoption request section clearly showed pending, approved, and rejected requests, reducing the chance of missing or duplicate work. The organized layout helped in handling adoption applications accurately, allowing staff to respond faster to user submissions. JavaScript-based form validations also helped avoid errors during the adoption form process, leading to better data quality and fewer incomplete submissions.

Functional Verification

Functional testing was done to make sure the main parts of the system work as intended: user login, managing pets, submitting adoption requests, and the admin approval process.

All Create, Read, Update, and Delete operations were performed successfully. The authentication part, using sessions or JWT, provided secure login options for both admins and users. Trying to access restricted areas without proper login was successfully blocked, showing that the security features worked well.

The adoption process was tested thoroughly.

Test users submitted adoption forms multiple times, and the system consistently stored these requests in the database without duplications or data issues. The admin review process worked smoothly, with each approval or rejection updating the status immediately and notifying users. The connections between pets, users, and adoption requests were set up correctly, allowing full tracking of each adoption request. Overall, the system met all the functional requirements from the design phase, indicating that the implemented software components are reliable.

Discussion

Overall, the results show that the Pet Adoption Website successfully meets its design goals by offering a fast, easy-to-use, and functional platform for managing pet adoptions.

The combination of a responsive frontend and a high-performance backend creates a smooth experience for both users and shelter administrators. While there are areas for improvement, the current system already shows strong potential for real-world use in animal shelters and pet rescue organizations. The scalability of Node.js and MongoDB positions the platform well for future growth, while the modular setup of Express.js allows for easy integration of new features as user needs change. The study highlights the value of using modern full-stack technologies for societal applications like pet adoption management.

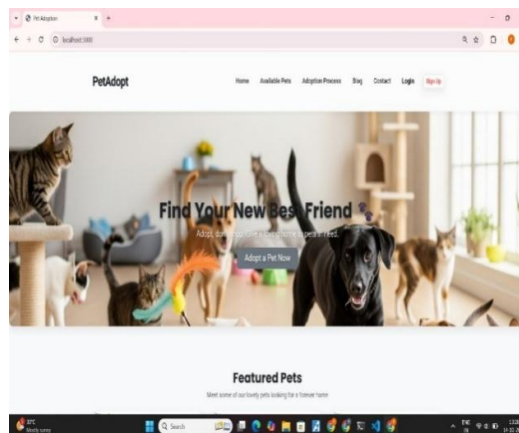


Fig-5 : Home page



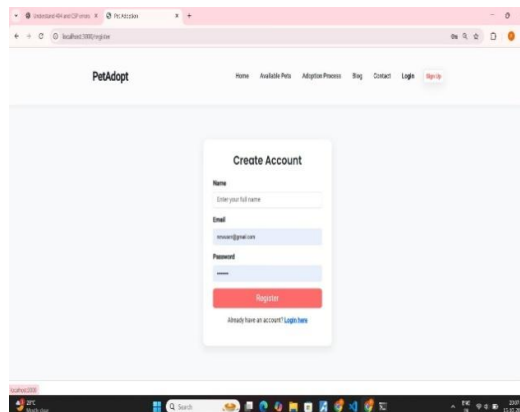


Fig-6: Create Account

VI. FUTURE SCOPE

The pet adoption management platform has great potential for growth and improvement, making it possible to turn it into a smarter, more user-friendly, and scalable system. One exciting future development is adding an AI-powered pet recommendation system. This system can use data like what users like, their past adoption choices, their lifestyle, and details about the pets to suggest the best matches for each person looking to adopt. This not only makes the experience better for users but also helps in making more successful and lasting adoptions. By using techniques such as collaborative filtering or content-based filtering, we can make sure that every visitor gets pet suggestions that match their interests and actions.

Another important feature to consider is adding a real-time communication tool, such as live chat or video calls.

This would allow people interested in adopting to talk directly with shelter workers, ask questions, discuss pet behaviour, and learn about the pet's health before they decide. Real-time communication helps reduce uncertainty and builds trust, which is really important when it comes to animal adoption. Using WebSocket-based chat systems or integrating third-party APIs for secure video calls can improve user interaction and make the whole process smoother.

To reach more people, the platform can be developed into a mobile app for both Android and iOS.

A mobile app lets users look at available pets, track their applications, get instant updates, and chat with shelters while on the move. Mobile apps offer features like customized settings, device notifications, and the ability to browse without an internet connection, all of which help create a better adoption experience. Shelters can also manage pet listings more easily with the app's camera and upload features, which helps them keep pet availability updated without delays.

A really innovative future feature is adding a GPS-based Lost & Found Pet Module.

This would let pet owners report if their pets are missing, and community members could help find them by using location services. Shelters can also work together to scan for microchip IDs and update the system in real time. Using GPS tracking through IoT tags or QR code collars could greatly improve the chances of finding lost animals, helping to ensure their safety and improve community well-being.

The platform can also be improved with a detailed vaccination and medical record system, allowing shelters to keep digital health profiles for each pet.

These records can include information on vaccinations, deworming, medical treatments, allergies, and surgical history. Giving adopters full access to these records helps them provide better care after adoption and supports veterinarians in giving timely treatment. Automating reminders for vaccinations for both shelters and adopters would also help keep pets healthy.

Finally, introducing a multilingual interface would make the platform more accessible to a wider range of people with different language backgrounds.



Supporting regional languages not only makes the platform more user-friendly but also helps increase adoption rates by allowing people from rural and semi-urban areas to use the platform more easily. This can be done through dynamic translation modules, localization libraries, and allowing users to choose their preferred language.

Overall, these future improvements—ranging from AI-powered recommendations to GPS tracking and multilingual support—will greatly enhance the platform's functionality, inclusiveness, and overall impact.

Adding these advanced features would turn the system into a complete digital environment for pet adoption, animal welfare, and community involvement.

VII. CONCLUSIONS

The Pet Adoption Website shows how modern web development can improve and bring digital tools into the pet adoption process. It combines an easy-to-use interface with a strong backend system, making it easier for people to look for pets, apply to adopt, and get approval from admins. The site handles tasks that used to be done manually, reducing the work for shelters and making the experience better for those looking to adopt. Features like updated pet listings, checks to make sure forms are correct, secure login systems, and real-time updates on adoption requests help make the whole process more organized, clear, and efficient.

When tested with sample data, the system proves it works well, is easy to use, and functions reliably. Using Node.js helps the server respond quickly, while MongoDB stores pet and user information in a flexible way. The website looks good on all devices thanks to Bootstrap and EJS, so anyone can use it without trouble. Admins also have a special dashboard that helps them manage requests, reduce mistakes, and make better decisions. All these things together show that the system achieves its goal of making the adoption process easier for everyone involved. More than just improving how adoption works, this project shows how digital platforms can support wider animal welfare efforts. By making adoption easier and more structured, the site encourages people to take better care of pets and helps shelters connect more effectively with the community. However, even though the platform does what it's meant to, there are still areas to grow. Adding features like AI suggestions, support for multiple languages, mobile apps, and a way to track lost pets could make the system even more powerful. These improvements would turn it into a complete tool for managing animal welfare.

In the end, the Pet Adoption Website is a great example of how technology can help solve real problems in society. It cuts down on manual work, makes data more accurate, makes things easier for users, and improves communication between shelters and adopters. With more development, the system could become a full-scale solution for managing adoption, keeping track of pets' health, helping the community, and improving overall pet welfare. This project shows how important digital changes are in helping social causes and sets the stage for future progress in animal care and adoption services.

VIII. ACKNOWLEDGEMENT

The successful completion of the Pet Adoption Website project is due to the consistent efforts, teamwork, and guidance of many people and organizations. The authors would like to deeply thank everyone who contributed, either directly or indirectly, to the research, development, evaluation, and improvement of this platform. This project wouldn't have been possible without the encouragement, knowledge, and helpful feedback received at every stage of its creation.

First and foremost, the authors want to express their sincere gratitude to their academic advisors and project supervisors.

Their ongoing support and expert guidance were crucial in shaping the direction, methods, and execution of this research. Their thoughtful suggestions, technical advice, and critical feedback greatly improved the depth and quality of the project. The authors are especially thankful for the time these mentors spent reviewing system parts, analyzing performance, and explaining complex technical topics like RESTful architecture, user login systems, and database optimization. Their mentorship encouraged a mindset of asking questions and solving problems methodically, helping the authors tackle technical challenges effectively.

The authors also want to thank the development team members who worked together throughout the design and build phases. Their hard work in writing clean code, fixing errors, connecting APIs, and improving the user interface and user



experience was essential in turning the project idea into a working software. The teamwork in areas like database structure, routing logic, session handling, and creating a responsive interface shows how important collaboration is in achieving technical excellence. Every team member's effort helped ensure the platform met high standards for performance, security, and usability.

The authors would also like to express their appreciation to the reviewers, testers, and peers who helped during the continuous evaluation process.

Their detailed feedback during early and later testing stages was very helpful in finding system issues, improving the user interface, and enhancing the overall experience. By simulating real-life user situations and testing different features—like user registration, browsing pets, adoption requests, and admin dashboard tools—these individuals strengthened the platform's reliability and stability. Their willingness to closely examine the system's features and offer useful suggestions made a big difference in refining the final version.

The authors also recognize the contributions of online tech communities, open-source developers, and documentation resources that supported the project.

Websites like Stack Overflow, GitHub, and guides from Node.js, Express.js, MongoDB, Bootstrap, EJS, and other open-source tools provided essential knowledge, code examples, solutions to problems, and best practices. These resources ensured the development process followed widely accepted coding standards and modern web development principles. The availability of such extensive knowledge reflects the spirit of collaboration in the global tech community, which made efficient development and problem-solving much easier.

The authors also thank the institutions and labs that provided the necessary tools and environment, including computers, software, reliable internet, and testing platforms. Access to development tools like Visual Studio Code, the Node.js environment, the npm package system, and database hosting allowed smooth work and consistent project progress. The academic setting that supported discussions and experiments played a key role in enabling the team to design, test, and improve the platform effectively.

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